UNIT - V
Serviceability & Staircases

Design of Reinforced Concrete Structures (23A01502) - handwritten study notes

1. Limit State of Serviceability

Introduction

- A structure must be **safe** (limit state of collapse) and also **useful** during its life (limit state of serviceability, LSS).
- Serviceability = the structure should not **deflect excessively, crack widely, or vibrate** so that appearance, comfort or function is spoiled.
- IS 456:2000 deals with serviceability in **clause 35** (deflection - 23.2, cracking - 35.3, durability - 8).
- Checks are made with **partial safety factor = 1.0** for loads (characteristic / working loads) and for materials.

Collapse check uses factored loads (1.5 x). Serviceability check uses working loads (1.0 x). Never mix the two.

Serviceability Limit States in IS 456

Limit state	Requirement	Clause
Deflection	Final deflection <= span/250	23.2
Deflection (partitions)	After construction <= span/350 or 20 mm	23.2 (b)
Cracking	Crack width <= 0.3 mm (0.2 mm severe)	35.3.2
Durability	Cover, w/c ratio, cement content	8, 26.4
Vibration	Avoid resonance with machines	35.1

- Check is done as **indirect** (span/depth ratios, bar spacing) or **direct** (calculation of deflection and crack width).

2. Deflection of RC Members

Limits as per IS 456 (clause 23.2)

- The **final** deflection (including creep and shrinkage) due to all loads <= **span / 250**.
- The deflection **after erection of partitions and finishes** <= **span / 350 or 20 mm**, whichever is less.

Two ways to satisfy the code: (1) **Indirect** - span / effective depth limits (simple, for design). (2) **Direct** - calculate deflection using Annex C (for checking).

Short Term Deflection

- Uses the elastic formula with an **effective moment of inertia** I_{eff} because the section is partly cracked.

$$E_c = 5000 \sqrt{f_{ck}} \ N/mm^2$$

$$\text{Short-term deflection } a_i = k\, w\, L^4 / (E_c\, I_{eff})$$

- $k = 5/384$ for simply supported beam with UDL; $k = 1/8$ for cantilever with UDL ($wL^4/8EI$).
- Effective moment of inertia (Annex C):

$$I_{eff} = I_r / \left(1.2 - (M_r/M)(z/d)(1 - x/d)(b_w/b)\right), \quad I_r <= I_{eff} <= I_{gr}$$

  - I_r = moment of inertia of the **cracked** section, I_{gr} = gross section.

Long Term Deflection (Creep and Shrinkage)

- Under sustained load, concrete **creeps**; deflection increases with time. Shrinkage curvature adds further deflection.
- **Creep deflection** is obtained using the **effective modulus** of concrete:

$$E_{ce} = E_c / (1 + \theta)$$

- θ = creep coefficient (depends on age at loading).

Age at loading	Creep coefficient theta
7 days	2.2
28 days	1.6
1 year	1.1

- Long term deflection = short term (permanent loads, using E_{ce}) + shrinkage deflection.

$$a_{cs} = k_3 \, psi_{cs} \, L^2$$

  - psi_{cs} = shrinkage curvature = k_4 $epsilon_{cs}$ / D, $epsilon_{cs}$ = **0.0003** (total shrinkage strain if no data).
  - k_3 = 0.125 for simply supported, 0.5 for cantilever. k_4 depends on (p_t - p_c) and p_t.

Final deflection $a = a_{cc} + a_{cs}$ (a_{cc} = creep + short term of permanent load)

- Total deflection is the sum of: **short term (total load)** + **creep (permanent load)** + **shrinkage**.

Example of E_{ce}: M20 concrete, Ec = 5000 sqrt(20) = 22,360 N/mm^2. At 28 days theta = 1.6, so E_{ce} = 22360 / 2.6 = 8,600 N/mm^2 (only 38% of Ec).

3. Deflection Control by Span / Depth Ratio

Basic Values (clause 23.2.1)

Support condition	Basic L / d
Cantilever	7
Simply supported	20
Continuous	26

- Valid for spans up to **10 m**. For spans above 10 m (except cantilevers)

multiply by **10 / span (m)**.

- For two-way slabs, basic values are applied on the **shorter span** (spans up to 3.5 m, f_y <= 415 N/mm^2).

$$(L/d)_{allowable} = (L/d)_{basic} \times k_t \times k_c \times k_f$$

- **k_t** = modification factor for **tension reinforcement** (Fig. 4)
- **k_c** = modification factor for **compression reinforcement** (Fig. 5)
- **k_f** = factor for flanged beams (Fig. 6 and 7) based on b_w / b_f.

Modification Factor for Tension Steel (Fig. 4)

- Depends on **percentage of steel** $p_t = 100\ A_{st} / (b\ d)$ and the **service stress** f_s in steel:

$$f_s = 0.58\ f_y\ (A_{st,req} / A_{st,prov})$$

- For Fe 415 with $A_{st,req} = A_{st,prov}$: $f_s = 0.58 \times 415 =$ **240 N/mm^2**. For Fe 500: $f_s =$ **290 N/mm^2**. For Fe 250: 145.
- **Maximum k_t = 2.0.** More steel (higher p_t) ⟶ smaller k_t. Less service stress ⟶ larger k_t.

Modification Factor for Compression Steel (Fig. 5)

- Depends on percentage of compression steel $p_c = 100\ A_{sc} / (b\ d)$. Max k_c = **1.5**.
- Compression steel reduces long term deflection. Typical: p_c = 0.25% gives about 1.08, p_c = 1.0% about 1.25.

Flanged Beams

- Flanged beams have more stiffness, so allowable L/d is **increased**: use Fig. 6 and 7 with the ratio **b_w / b_f**.
- Rule of thumb: for b_w/b_f = 0.3 the factor is about **0.8** on the basic value for p_t calculation. Interpolate for ratios between 0.3 and 1.0.
- The percentage of steel p_t for flanged beams is based on the **web width b_w**.

Worked Example 1 - Deflection check of a beam

Given: Simply supported RC beam, effective span 6 m, b = 230 mm, d = 450 mm. Provided 4 bars of 20 mm dia ($A_{st,prov}$ = 1256 mm^2), $A_{st,req}$ = 1100 mm^2. Provided 2 bars of 16 mm in compression (A_{sc} = 402 mm^2). Fe 415, M20. **Check deflection.**

- Basic L/d = **20** (simply supported, span < 10 m).
- p_t = 100 x 1256 / (230 x 450) = **1.21 %**
- f_s = 0.58 x 415 x (1100 / 1256) = 240.7 x 0.876 = **211 N/mm^2**
- From Fig. 4: for p_t = 1.21 % and f_s = 211, **k_t = 1.1** (approx.)
- p_c = 100 x 402 / (230 x 450) = 0.39 %. From Fig. 5, **k_c = 1.1**
- Allowable L/d = 20 x 1.1 x 1.1 = **24.2**
- Actual L/d = 6000 / 450 = **13.3** < 24.2 hence **safe in deflection**.

If actual L/d is more than allowable: increase depth d, or increase steel (reduces f_s), or provide compression steel. Redesign and re-check.

Worked Example 2 - Minimum depth from deflection

Given: Cantilever slab with projection 1.5 m, M20, Fe 415, assumed k_t = 1.4 (p_t about 0.3 %).

- Basic L/d for cantilever = 7; allowable L/d = 7 x 1.4 = **9.8**
- d >= 1500 / 9.8 = **153 mm**. Provide d = 155 mm, D = 180 mm (cover 20 + 5).
- This is the **depth for stiffness**, which is checked later for strength (Mu,lim).

4. Cracking in RC Members

Why do cracks form?

- Concrete is weak in tension (about 10% of compressive strength): it cracks when the tensile strain exceeds about **0.0001 to 0.0002**.
- Cracks are **normal** in RC; what we control is **width** (appearance, durability, corrosion of steel).

Causes and Types of Cracks

Type	Cause	Time
Plastic shrinkage	Rapid evaporation of surface water	First hours
Plastic settlement	Settlement of fresh concrete over bars / formwork	1-3 hours
Early thermal	Heat of hydration, cooling	1 day - 2 weeks
Drying shrinkage	Loss of moisture, restrained	Weeks / months
Flexural cracks	Tension due to bending (vertical cracks)	Under load
Shear / diagonal	Principal tension near supports	Under load
Corrosion cracks	Rusting expands steel (7 x volume)	Years
Bond / splitting	Poor anchorage along bars	Under load

Crack Width Limits (IS 456 clause 35.3.2)

- **0.3 mm** at the tension face for members in **normal exposure** (mild, moderate).
- **0.2 mm** for members in **severe / aggressive exposure** (sea water, chemicals) and **0.1 mm** for water-retaining structures (IS 3370).
- Crack width is measured at the level of steel surface nearest to the tension face.

Control of Cracking: Indirect Method

Cracks are controlled by detailing rules (clause 26.3.3) without calculating crack width:

- **Maximum spacing** of tension bars: **clear distance** between bars should not exceed the values in **Table 15** (depend on f_y and moment redistribution).

f_y (N/mm^2)	Max clear distance, 0% redistribution
250	300 mm
415	180 mm
500	150 mm

- Distance reduces for **negative redistribution** (e.g. -30%: 215 / 125 / 105 mm for Fe 250 / 415 / 500).
- **Side face reinforcement**: if depth of web **D > 750 mm**, provide 0.1% of web area (b_w x D) distributed equally on both faces, spacing <= **300 mm or web thickness, whichever is less**.
- **Minimum bar spacing**: greater of bar diameter and (maximum size of aggregate + 5 mm). Minimum cover as per exposure.

Rule: use **many small bars** instead of few big bars: better crack distribution and bond.

Direct Method: Crack Width Calculation (Annex F)

$$w_{cr} = 3\, a_{cr}\, \text{epsilon}_m / (1 + 2\,(a_{cr} - c_{min}) / (h - x))$$

- a_{cr} = distance from the point considered to the surface of the nearest longitudinal bar.
- c_{min} = minimum clear cover to the longitudinal bar.
- **h** = overall depth of the member, **x** = depth of neutral axis (cracked section).
- **epsilon_m** = average strain at the level considered, obtained after deducting **tension stiffening** of concrete from the steel strain.

Measures to Control Cracking

- Provide **adequate cover** (but not excessive) and use closely spaced small bars.
- Reduce **steel stress** (use more steel or bigger section).

Serviceability summary: Deflection by L/d x modification factors. Cracking by bar spacing (Table 15), side face steel for D > 750 mm, w_{cr} <= 0.3 mm.

5. Staircases: Introduction

- A **staircase** is a sequence of steps that provides access between floors. A stair slab is designed like a **one way slab** (inclined).

Terms used in Stairs

- **Tread (T)**: horizontal part of a step on which foot rests. **Going**: horizontal distance covered by the whole flight (no. of treads x tread).
- **Rise (R)**: vertical height of a step. **Riser**: vertical face of the step.
- **Waist**: thickness of the slab (measured perpendicular to the slope).
- **Flight**: a series of steps between two landings. **Landing**: platform at the end of a flight, for rest and change of direction.

Proportions of Steps

- **Rise R = 150 to 190 mm** (residential 150-175; public buildings 150). **Tread T = 250 to 300 mm**.
- Thumb rule: **2R + T = 550 to 700 mm** (or R + T = about 400 mm).
- Width of stair: residential **1.0 to 1.25 m**; public **1.5 to 2.0 m**; maximum 12-15 risers in one flight.
- Slope of stair: 25 to 40 degrees (normal 30 to 35 degrees).

Types of Staircases

- **Straight stair**: single flight / several flights in one line. Simple; needs long space.
- **Quarter-turn stair**: 90 degrees turn through a landing (L shaped).
- **Half-turn / Dog-legged stair**: two flights in opposite directions with a **mid landing**; **no well**, most common, compact.
- **Open well stair**: two or more flights around a **well (opening)** between them; used in public buildings.
- **Geometrical stair**: open well with **curved** turn, no newel (the handrail is a smooth curve).
- **Spiral stair**: steps radiate from a central post / column; takes least floor space, used for fire escape or limited access.
- **Helicoidal stair**: slab twisted in helix shape without central support;

torsion governs; costly.

> Dog-legged stairs are most popular for residential buildings and design problems.

6. Loads on Staircases

Dead Load

- **Self weight of waist slab on slope** = 25 x D (kN/m² of slope). Convert to horizontal plan area:

$$w_s = 25\ D \times \sqrt{R^2 + T^2} / T \ (kN/m^2 \text{ of plan})$$

- **Weight of steps** (triangular): $w_{step} = 25 \times R / 2$ (kN/m² on going).
- **Finishes**: 0.6 to 1.0 kN/m² (tiles, plaster). Landing: 25 x D + finishes.

Live Load (IS 875 Part 2)

Building	LL on stairs
Residential, offices (not crowded)	3.0 kN/m²
Public buildings, assembly halls	5.0 kN/m²
Where stairs may be overcrowded	5.0 kN/m²

- Live load acts on the **horizontal projection**. Handrail load 0.75 kN/m as per IS 875.
- **Design load** $w_u = 1.5\ (DL + LL)$.

Effective Span of Stairs (clause 33.1)

- **Stairs spanning longitudinally** (supported at the ends of flight): effective span = c/c of supports, or clear span + effective depth, **whichever is less**.
- If the **landing slab spans in the same direction** as the flight: effective span = **going + the width of the landing** supports (c/c).
- If the **landing spans at right angles** to the flight: effective span = going + at each end **half width of landing or 1.0 m**, whichever is

smaller.

Staircase Spanning Longitudinally vs Transversely

Longitudinal (along the flight)	Transverse (across the width)
Slab supported at bottom and top landing beams	Supported on side walls or stringer beams
Span = going + landings	Span = width of stair (small)
Steps carried by waist slab	Steps act as beams / waist is thin
Used for dog-legged, common	Used for cantilever steps from a wall, open stairs
Main steel along slope	Main steel along width

Whenever possible provide the support to the landing on **both** sides; this lowers the span, cost and deflection.

7. Design of a Dog-Legged Staircase

Design Steps

Fix R, T, no. of steps → Effective span L → Assume depth (L/d x MF) → Calculate loads (going, landing)

BM and SF

Check d from Mu,lim → Find A_{st} → Distribution steel → Check shear, deflection

Check development length, detailing

Points to Remember

- Slab is designed per metre **width** of the stair. Loads are on **horizontal plan** area.
- Bending moment (simply supported): $M = w L^2 / 8$. When the slab is continuous with the landing/beams, take $w L^2 / 10$ (as per IS 456 Annex 22.5, end spans) for midspan. For stairs built into walls, $w L^2 / 10$ is also used.
- For stairs spanning with landing in the same direction: calculate loads separately on flight and landing, find reactions and **maximum BM** at the section of zero shear.
- Depth of waist slab: $L / 20$ **to** $L / 25$ for simply supported; $L / 26$ **to** $L / 30$ with continuity.
- Main steel along the slope. **Distribution steel**: 0.12% of gross area (HYSD), perpendicular to main bars.
- At the junction of flight and landing in dog-legged stairs, **provide extra bars** (re-entrant corner bars) and tie so that the bars do not straighten and burst the concrete.

Worked Example 3: Design of a Dog-Legged Staircase

Given: Floor to floor height **3.2 m**. Two equal flights, stair width **1.2 m**, landing width 1.2 m. Live load **3 kN/m²**, floor finish 1.0 kN/m². **M20 and Fe 415.** Landing slab spans in the same direction, resting on 230 mm walls. Design the flight.

Step 1: Proportions

- Height of one flight = 3.2 / 2 = **1.6 m**. Rise R = 160 mm, so number of risers = 1600 / 160 = **10**.
- Treads per flight = 10 - 1 = **9**; tread T = **250 mm**; going = 9 x 0.25 = **2.25 m**.

Step 2: Effective span

- Flight (going) = 2.25 m, plus landing 1.2 m in the same direction.
- Effective span L = 2.25 + 1.2 = **3.45 m** (c/c of supports, taking the

walls as simple supports).

Step 3: Trial depth

- Take L/d = 20 x 1.5 (modification) = 30, so d = 3450 / 30 = 115 mm. **Take D = 170 mm, d = 145 mm** (cover 20 mm + bar 12/2 = 5 mm; D = 145 + 25).

Step 4: Loads (per m width)

Item	Calculation	kN/m²
Waist slab on slope	25 x 0.17 x sqrt(0.16²+0.25²)/0.25 = 4.25 x 1.187	5.05
Steps	25 x 0.16 / 2	2.00
Finishes	given	1.00
Live load	given	3.00
Total on going w_1	DL + LL	11.05
Landing: self weight	25 x 0.17	4.25
Landing: finish + LL	1.0 + 3.0	4.00
Total on landing w_2		8.25

- Factored loads: going **w_{u1} = 1.5 x 11.05 = 16.58 kN/m**; landing **w_{u2} = 1.5 x 8.25 = 12.38 kN/m**.

Step 5: Bending moment and shear

- Load on going = 16.58 x 2.25 = 37.30 kN (acts at 1.125 m from A); load on landing = 12.38 x 1.2 = 14.86 kN (acts at 2.85 m from A).
- Taking moments about A: R_B = (37.30 x 1.125 + 14.86 x 2.85) / 3.45 = **24.44 kN**; R_A = 52.16 - 24.44 = **27.72 kN**.
- Zero shear at x = 27.72 / 16.58 = **1.672 m** from A (lies in the going).
- **M_u = 27.72 x 1.672 - 16.58 x 1.672² / 2 = 46.35 - 23.18 = 23.17 kN.m per m**.
- Check with w L^2 / 8: average w = 52.16/3.45 = 15.12; M = 15.12 x 3.45² / 8 = 22.5 kN.m (close).
- Design shear V_u = **27.72 kN**.

Step 6: Check depth

- $M_{u,lim} = 0.138\ f_{ck}\ b\ d^2 = 0.138 \times 20 \times 1000 \times 145^2 =$ **58.0 kN.m** > 23.17 kN.m, so section is under-reinforced. **d is OK**.

Step 7: Main steel

$$M_u / (b\ d^2) = 23.17 \times 10^6 / (1000 \times 145^2) = 1.10\ N/mm^2$$

$$A_{st} = 0.5\ (f_{ck}/f_y)\ [\ 1 - sqrt(1 - 4.6\ M_u/(f_{ck}\ b\ d^2))\ ]\ b\ d = 476\ mm^2$$

- Minimum steel = 0.12% x 1000 x 170 = 204 mm^2 < 476 mm^2, OK.
- Spacing of 12 mm bars = 1000 x 113 / 476 = 237 mm. **Provide 12 mm dia @ 200 mm c/c** ($A_{st,prov}$ = 565 mm^2); max spacing 3d = 435 or 300 mm, OK.

Step 8: Distribution steel

- A_{st} = 0.12% x 1000 x 170 = **204 mm^2**. Using 8 mm bars: spacing = 1000 x 50.3 / 204 = 246 mm. **Provide 8 mm @ 240 mm c/c** (max Sd = 725 or 450 mm).

Step 9: Check for shear

- $tau_v = V_u / (b\ d)$ = 27720 / (1000 x 145) = **0.19 N/mm^2**.
- p_t = 100 x 565 / (1000 x 145) = 0.39 %, so tau_c = 0.43 N/mm^2 (Table 19, M20). For slab thickness 170 mm, k = 1.26, so k tau_c = **0.54** > 0.19. **Safe in shear** (no shear steel).

Step 10: Check for deflection

- Basic L/d = 20. f_s = 0.58 x 415 x (476/565) = **203 N/mm^2**. From Fig. 4 with p_t = 0.39: k_t = **1.5**.
- Allowable L/d = 20 x 1.5 = **30**. Actual L/d = 3450 / 145 = **23.8** < 30. **Safe in deflection.**

Step 11: Development length

- $L_d = 0.87\ f_y\ phi / (4\ tau_{bd})$; tau_{bd} = 1.2 x 1.6 = 1.92 N/mm^2 (deformed bars). L_d = 0.87 x 415 x 12 / (4 x 1.92) = **564 mm (47 phi)**. Bars must be extended at least this length into supports.

Detailing

- Main bars: 12 mm @ 200 mm c/c along the slope, bent up at the landing. Provide **top steel (0.5 of bottom)** at supports for continuity: 10 mm @ 250 mm c/c for length L/5 to L/4.
- Distribution bars: 8 mm @ 240 mm c/c on top of main bars. Clear cover 20 mm.
- At the junction of the two flights, the bars should be bent properly and **cross each other**, not be continuous in a straight line.

Key point: Check list for stairs - rise/tread, effective span, loads on slope converted to plan, $M = wL^2/8$ or max BM by reactions, d from Mu,lim, Ast, distribution steel (0.12%), shear ($tau_v < k\ tau_c$), deflection (L/d), Ld.

8. Revision Notes

- **Serviceability** = deflection + cracking + durability + vibration. Checks use **working loads**.
- Deflection limits: **L/250 total**; **L/350 or 20 mm** after partitions.
- Basic **L/d** = 7 (cantilever), 20 (simply supported), 26 (continuous). Multiply by k_t, k_c, k_f. Reduce by 10/span for span > 10 m.
- **$f_s = 0.58\ f_y\ (A_{st,req}/A_{st,prov})$**. $E_{ce} = Ec / (1 + theta)$. Ec = 5000 sqrt($f_{ck}$).
- Cracking: **0.3 mm**, Table 15 bar spacing, side face steel if D > 750 mm (0.1% of web area), Annex F crack width formula.

Important Questions

- 1. What are the limit states of serviceability? Explain the serviceability requirements of IS 456.
- 2. How is deflection controlled in RC beams? Explain the modification factors for tension and compression steel.
- 3. Explain short term and long term deflections. What is the effect of

creep and shrinkage?

- 4. A simply supported beam of span 6 m has b = 230, d = 450, 4 - 20 mm bars. Check the deflection.
- 5. What are the causes of cracks in RC members? Explain the methods to control cracks.
- 6. State the clause 26.3.3 rules for the spacing of reinforcement. What is the side face reinforcement?
- 7. Explain the different types of staircases with neat sketches.
- 8. Design a dog-legged staircase for a floor height of 3.2 m. Live load 3 kN/m^2, M20, Fe 415. Draw the reinforcement details.